IMPLEMENTATION REFERENCE — HARLEYS ODOO 19 ENTERPRISE

# Batch/Lot Tracking on the Operations Grid

How Batch, Batch Qty and Expiry Date work on the Inventory Operations grid across Indent, Inventory, Purchase and Sales — one shared screen, four business flows, and only one place a brand-new batch is ever minted.

**Module** harleys_customization (view + JS/SCSS only) **Status** Working; §7 governance requirement not yet built

**Scope** stock.picking / stock.move / stock.move.line / stock.lot

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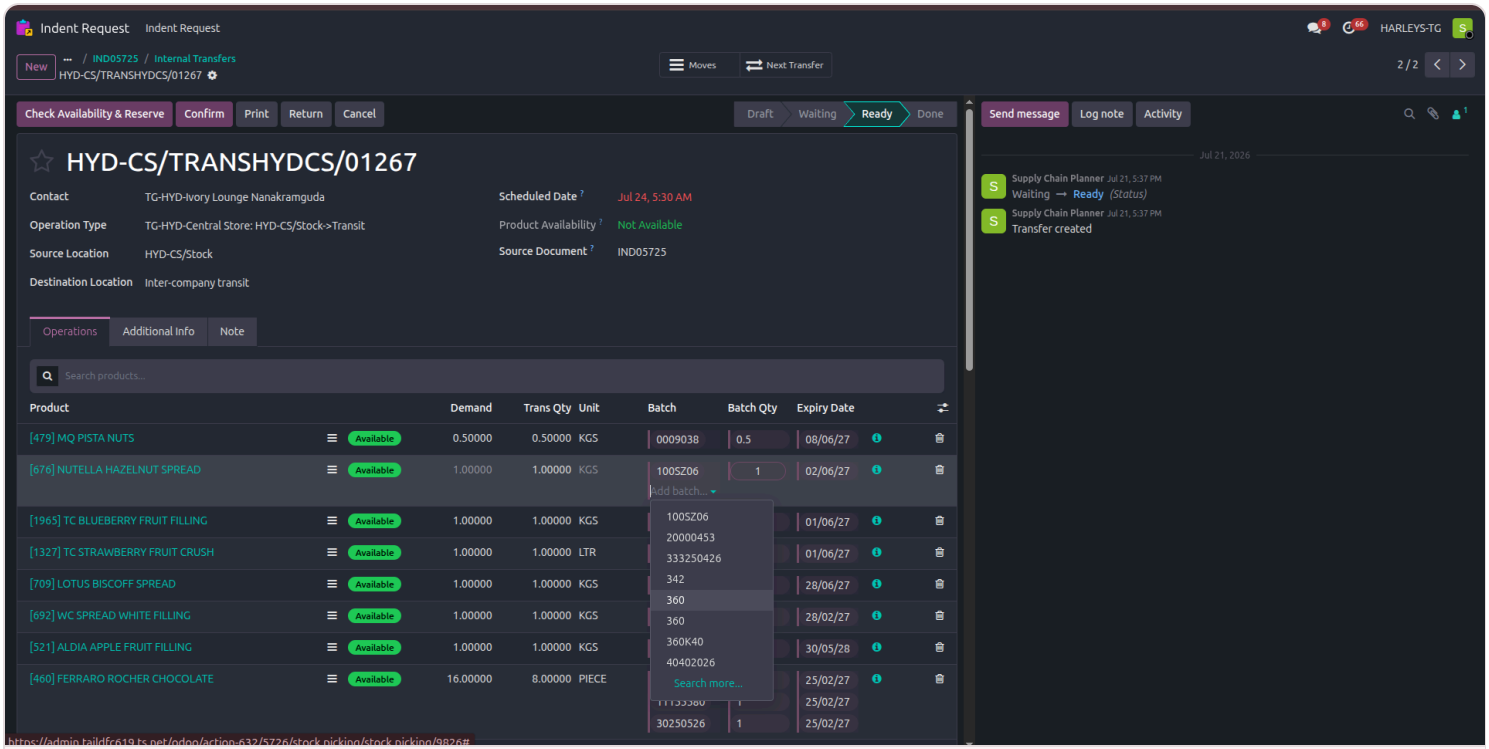
## 1. One screen, four business flows

Indent, Inventory, Purchase and Sales don't have four separate Operations screens — they all render through the *same* Odoo form, `stock.view_picking_form`, extended once by `harleys_customization/views/stock_picking_views.xml` (`view_picking_form_custom`). That inherited view isn't scoped to any specific action or menu, so every picking gets the same Operations-tab extensions regardless of how it was created. The four flows only diverge at the *field* level, via `picking_type_code` on `stock.picking` (`picking_code` — the same thing — related onto `stock.move`):

| BUSINESS FLOW | PICKING_TYPE_CODE | ENTRY POINT |
|---|---|---|
| Indent Request → Internal Transfer | `internal` | Indent Request form → "Internal Transfer" |
| Inventory → Internal Transfer | `internal` | Inventory app → Internal Transfers |
| Purchase → Receipt (GRN) | `incoming` | Purchase Order → Receipt |
| Sales → Delivery | `outgoing` | Sales Order → Delivery |

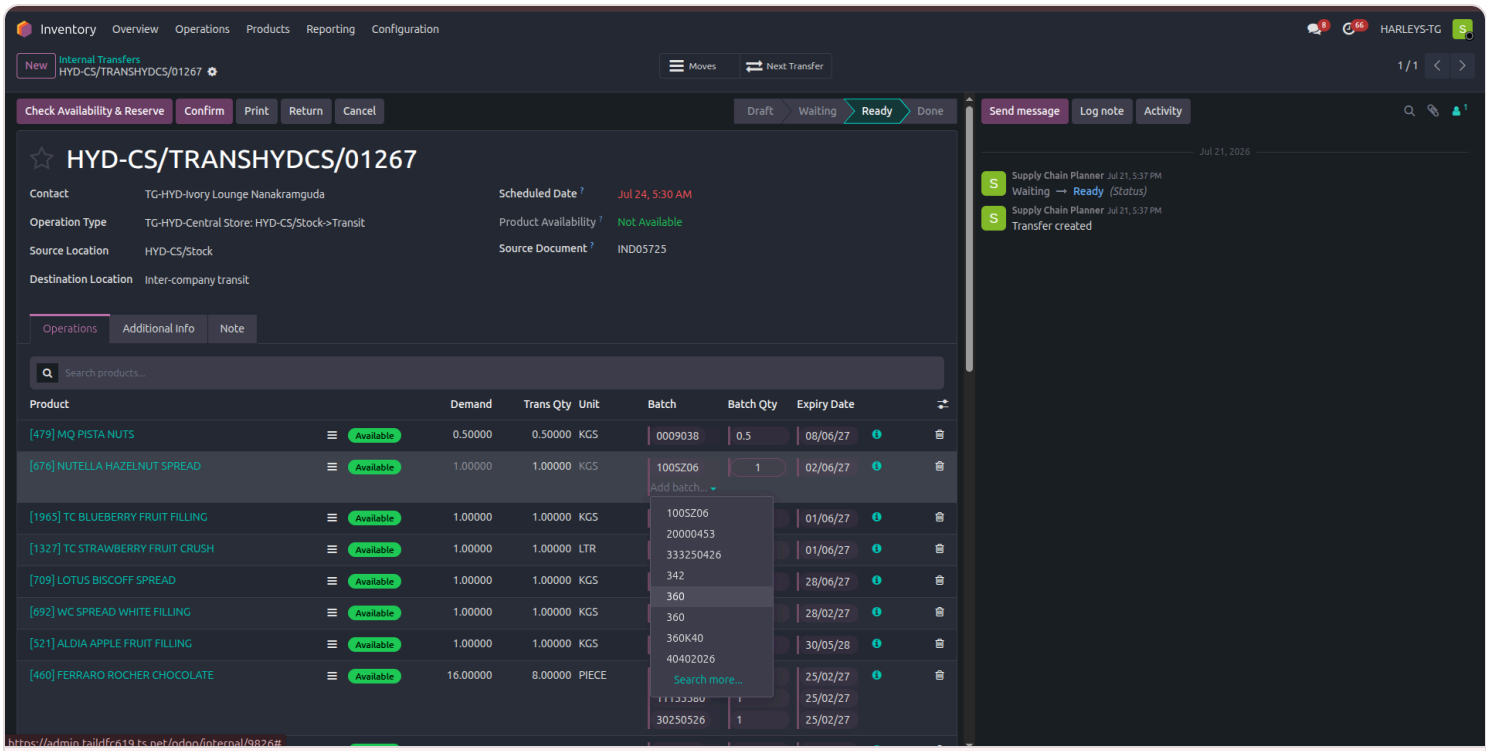
> **INDENT AND INVENTORY ARE LITERALLY INDISTINGUISHABLE BY `PICKING_TYPE_CODE`**
>
> Both are `internal`. The only way to tell them apart in the data is whether a `stock.picking` is linked from an `indent.request` record — the Operations grid itself treats them identically, as the two screenshots below show for the exact same underlying record.

## 1 Indent Request → Internal Transfer




**indent_internal_transfers_receipt.png** — `HYD-CS/TRANSHYDCS/01267`. The "Add batch…" dropdown lists only existing batches (100SZ06, 20000453, 333250426, 342, 360, …) plus "Search more…" — no create option, because this is an `internal` picking.

### 2 The same record, opened from Inventory → Internal Transfers




**inventory_internal_transfers.png** — same picking record as above, reached through a completely different menu. The Operations grid and its Batch dropdown behave identically either way, because both are the same `internal` picking type underneath.

## 2. The three Batch columns

| COLUMN | FIELD | WIDGET | EDITABLE? |
|---|---|---|---|
| Batch | `lot_ids` | `stock_move_line_batch` | Add/select/delete |
| Batch Qty | `move_line_ids` (`quantity`) | `stock_move_line_row_field`, `options="{'row_field': 'quantity', 'editable': True}"` | Yes, numeric input |
| Expiry Date | `move_line_ids` (`expiration_date`) | `stock_move_line_row_field`, `options="{'row_field': 'expiration_date', 'date_format': 'dd/MM/yy'}"` | Read-only, `dd/MM/yy` |

All three are `column_invisible` unless `picking_type_code` is one of `internal` / `incoming` / `outgoing` — specific to these four flows and absent from any other picking type Harleys might use.

All three read from one shared row list — `getMoveLineRows()` in `static/src/js/stock_move_line_rows.js` — built primarily off `lot_ids.records`, with a fallback for Receipt lines whose quantity is still 0 (Harleys forces new GRN lines to `quantity = 0`, which would otherwise make core's own `_compute_lot_ids` hide the row before the user can type a quantity). Because all three columns share this one function, adding or deleting a batch in one column is reflected in the other two with no separate bookkeeping.

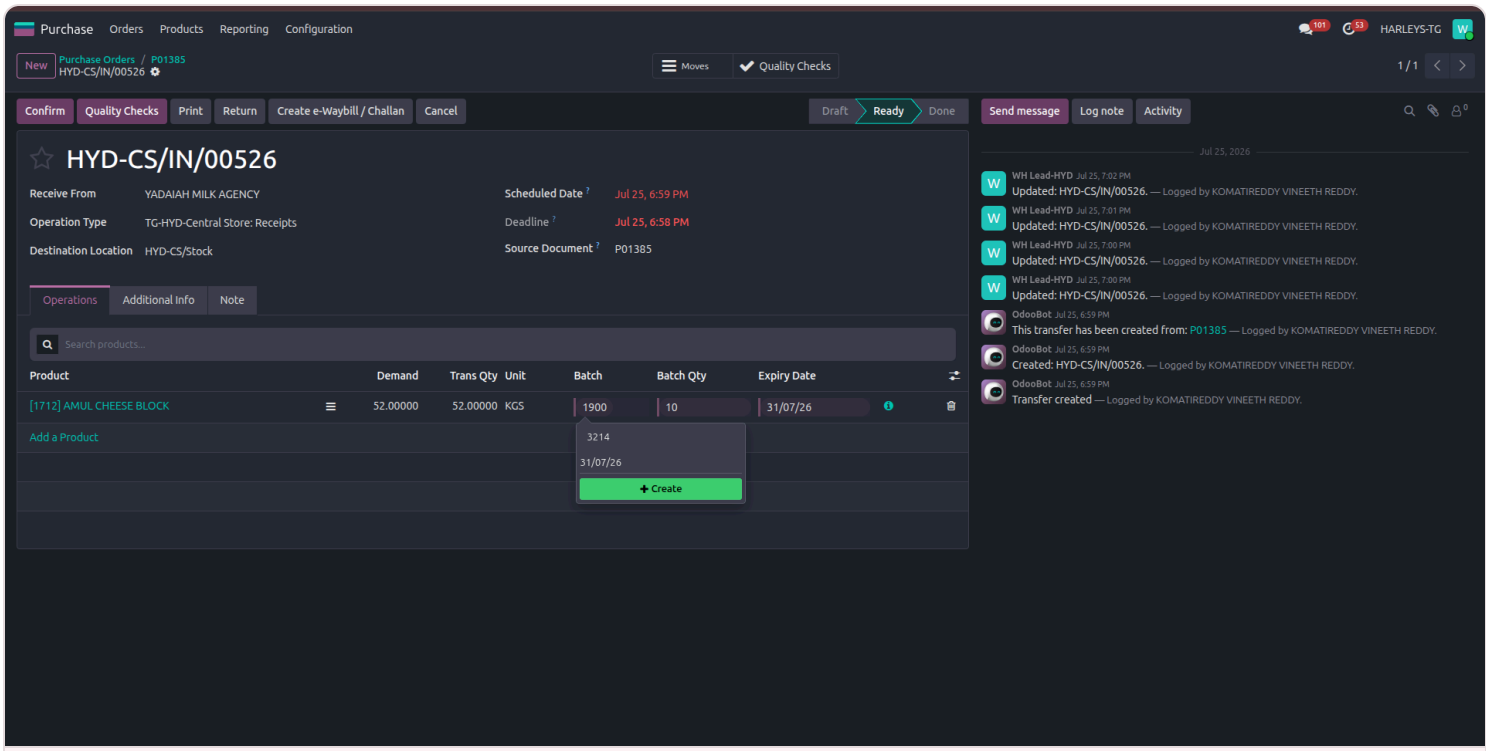
## 3. Where batches can be created vs only selected

This is the key difference between the four flows:

- **Purchase GRN (`incoming`) only** — clicking "Add" opens a dedicated popover (`StockMoveLineBatchQuickCreate`): live search against existing `stock.lot` records for the current product, and the ability to type a brand-new batch number. If the product tracks expiry, an expiry date is mandatory, entered via Odoo's own calendar widget locked to `dd/MM/yy` and unable to be backdated. **This is the only place in the system a new batch/lot record is minted through this grid.**
- **Indent/Inventory internal transfer and Sales delivery (`internal`, `outgoing`)** — clicking "Add" opens a plain inline autocomplete (`Many2XAutocomplete`), domain-filtered to the line's own product (`domain="[('product_id','=',product_id)]"`). Select-only in practice: there's no dedicated "new batch" UI here the way the Receipt popover has one. (The underlying widget still technically supports create-by-typing if `use_create_lots` is on for that picking type and the view's `options="{'create': [...]}"` allows it — that's bare Odoo default behavior inherited from `Many2XAutocomplete`, not a designed create flow, and isn't how these three flows are meant to be used.)

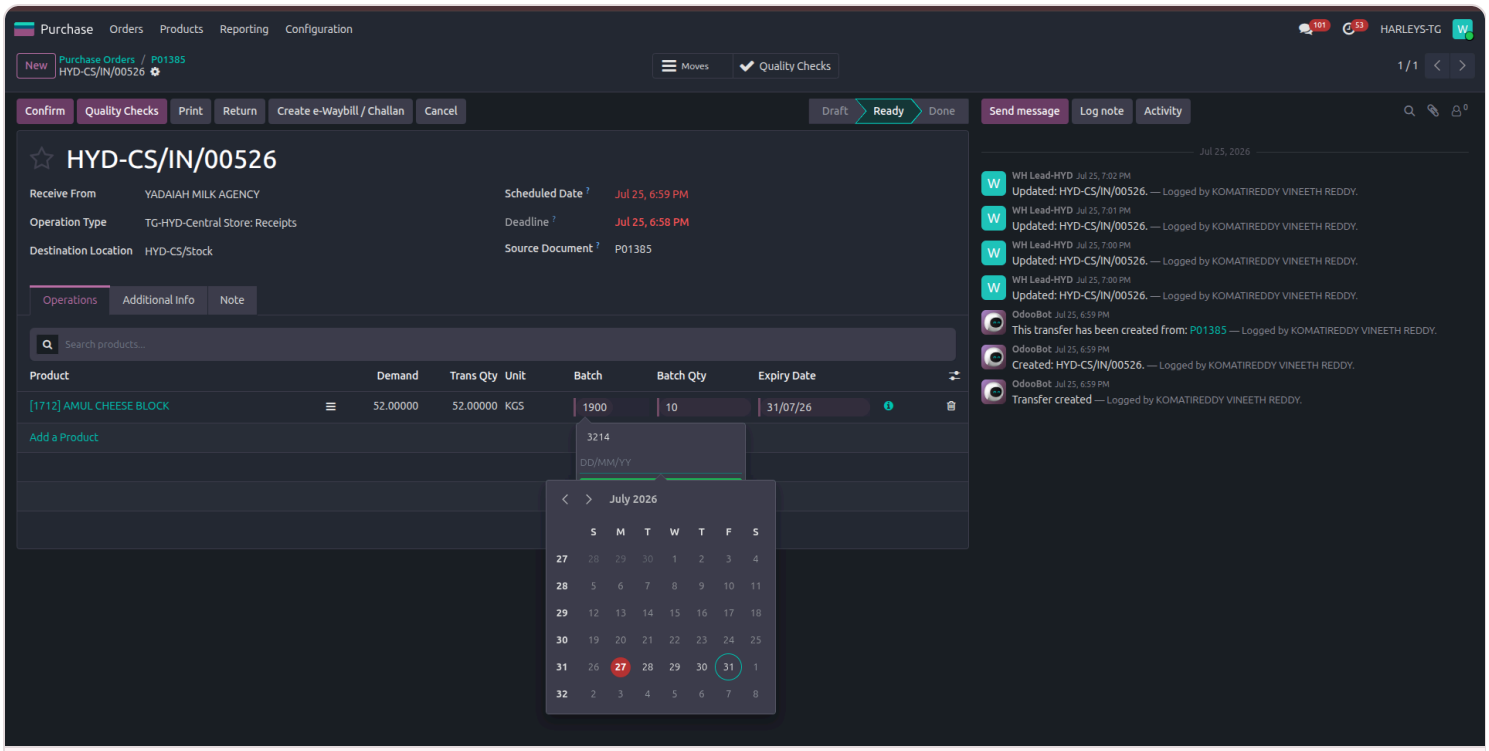
This matches how the business actually works: batches physically originate at goods receipt, so GRN is the only point that should mint new ones — the other three flows just draw down against what already exists.

## 1 Receipt — an existing batch, with a "+ Create" option alongside it



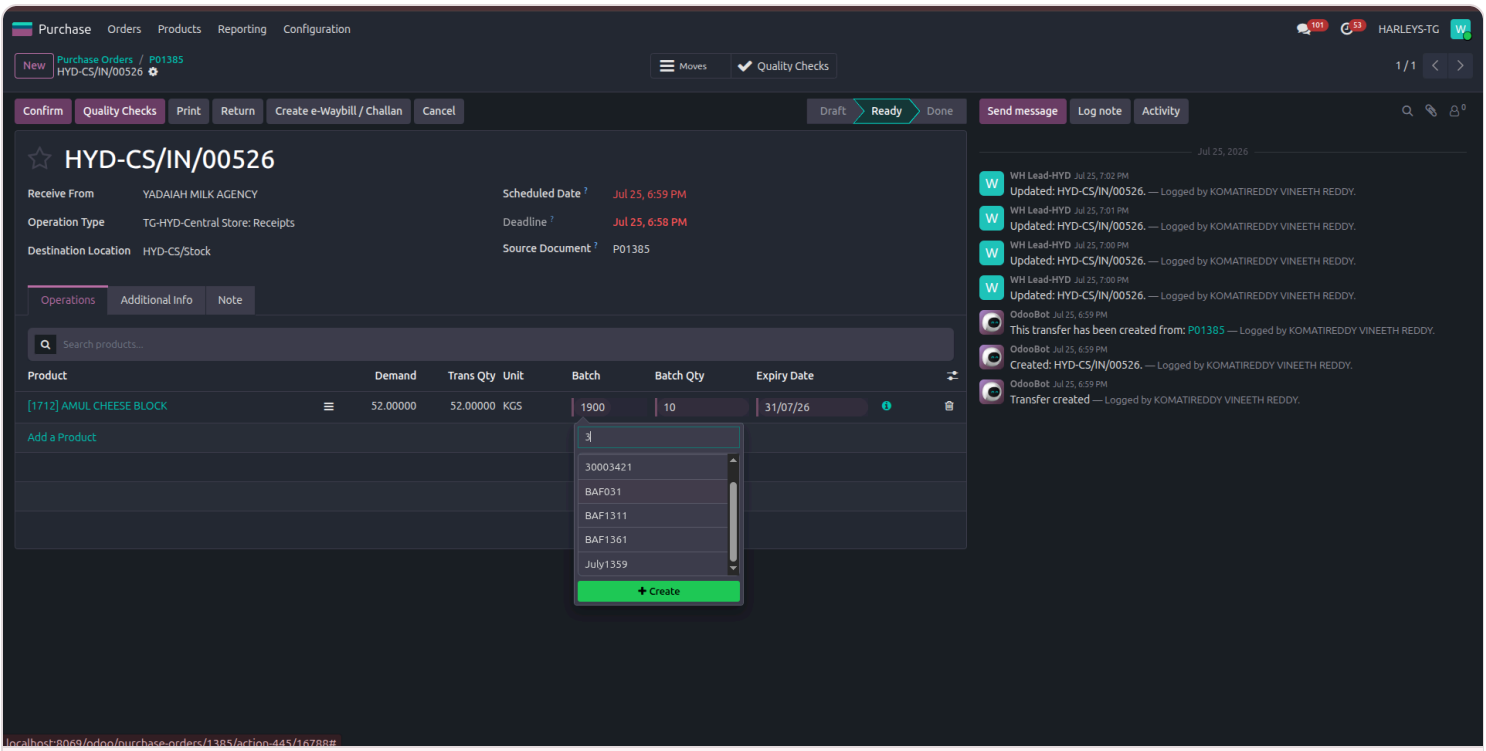

**purchase_orders_create_new_batch.png** — `HYD-CS/IN/00526`, product **AMUL CHEESE BLOCK**. The popover lists existing batch 3214 (dated 31/07/26) *and* a "+ Create" button — the option that doesn't exist on the other three flows.

## 2 Receipt — expiry date is mandatory, no backdating



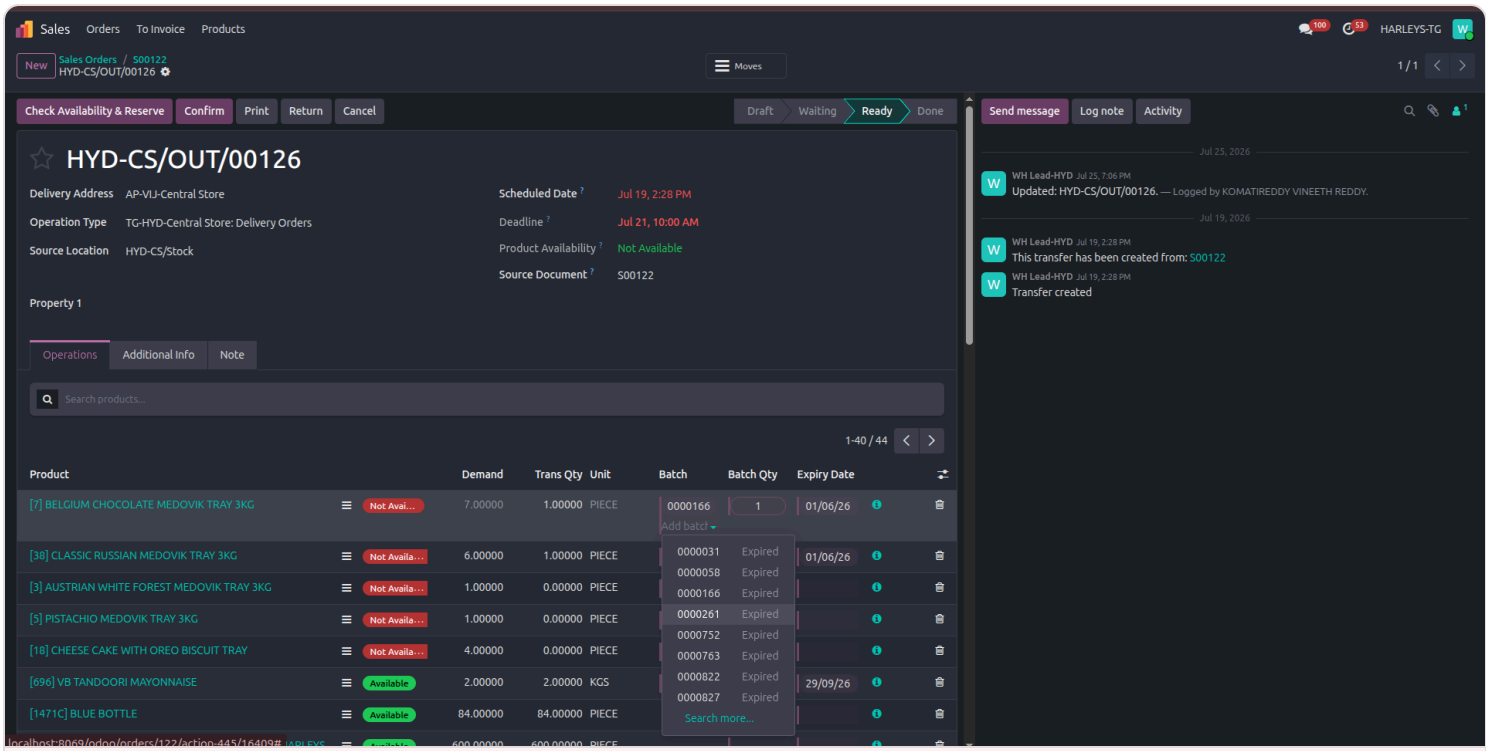

**purchase_orders_create_new_batch_select_exp_date.png** — same Receipt. The calendar widget is locked to `DD/MM/YY` and every date before today (the 27th) is greyed out and unselectable, so a new batch's expiry can never be backdated.

## 3 Receipt — live search and create share the same popover




**purchase_orders_select_if_batch_exists.png** — typing "3" surfaces matching existing batches (30003421, BAF031, BAF1311, BAF1361, July1359) in the same popover as the "+ Create" button, so a receiving clerk can pick an existing batch or mint a new one from one field, without switching screens.

### 4 Delivery — select-only, same as the internal flows




**sales_order_delivery.png** — `HYD-CS/OUT/00126`, product **BELGIUM CHOCOLATE MEDOVIK TRAY 3KG**. Like the two internal-transfer screenshots in §1, the dropdown only lists existing batches — no "+ Create" here, confirming Delivery is select-only exactly like Indent/Inventory. Note that every batch listed for this product is marked "Expired" and all remain selectable from this dropdown; nothing in the current view filters expired batches out.

## 4. Universality — a batch created once is visible everywhere

`stock.lot` is master data, not scoped to the picking that created it. The moment a batch is created on a GRN receipt, it's a normal `stock.lot` row (scoped only by `product_id` / `company_id`) and is immediately selectable from the same product-filtered dropdown on any other picking — internal transfer or delivery — for that product, company-wide. No extra work was needed for this; it falls directly out of `stock.lot` already being one shared model.

> **NO WAREHOUSE/LOCATION SCOPING ON THE LOT RECORD ITSELF**
>
> Availability *by location* is a separate concern, tracked on `stock.quant`, not gated on the lot. A batch created at one warehouse is, today, selectable from the Batch dropdown at any other warehouse for the same product — see §8.

## 5. Line-item batch add/delete — currently unrestricted

On all four flows, any user who can see the picking and has Odoo's own `stock.group_production_lot` (the base "Lots & Serial Numbers" feature toggle — a feature switch, not a rank/hierarchy group) can freely add or delete Batch rows on any line item (`deleteLot()` / `deleteLine()` in `stock_move_line_batch_field.js`). The only other gate is the standard `is_locked` / `state` check already on the whole Operations tab — batch add/delete disappears once a picking is done and locked, same as the rest of the grid. There is no separate, tighter permission for this action.

## 6. Not yet built: "only a higher official can change batch details"

> **STATED REQUIREMENT, NOT CURRENT BEHAVIOR**
>
> Flagging this clearly so it isn't documented as already-shipped. Checked directly against the code (`security/ir.model.access.csv`, `security/security_group.xml`): there is no ACL or `ir.rule` on `stock.lot` in this module at all.

Concretely, today:

- Editing an *existing* batch's own fields (name, expiration date, etc.) — via core's "Details" popup (`action_show_details` on `stock.move`, core Odoo) or the standalone Inventory → Products → Lots/Serial Numbers list — is open to anyone who can see the picking or that menu. No extra permission check exists beyond ordinary `stock.lot` access.
- There's already a precedent pattern in this same module for exactly this kind of split, just not wired to batches: `group_indent_admin` / `group_indent_user` (`res.groups.privilege` "Indent Request", `security/security_group.xml`), paired with `ir.rule` domain restrictions (e.g. `indent_request_rule` scoping non-admins to `requested_by = user.id`, `indent_request_manager_rule` giving the admin group unrestricted access). That's the established "user can touch their own records, admin/manager can touch everything" template in this codebase — the natural one to extend for batch-detail editing, as an alternative to reusing core's existing `stock.group_stock_manager` if "higher official" is meant to map to Inventory Manager specifically rather than a new bespoke group.

Before this can be built, needs a decision on: which group/role counts as the "higher official" (existing `stock.group_stock_manager`? a new group? something per picking type / department?), and exactly what "change batch details" covers — editing the `stock.lot` record's own fields after creation, versus something else

like reassigning which lot a saved move line points to. Line-item add/select/delete (§5) is explicitly *not* meant to be restricted by this — only editing a batch's own details after it exists.

## 7. Module reference — `harleys_customization`

| FILE | PURPOSE |
|---|---|
| `views/stock_picking_views.xml` | Adds the three columns to `stock.view_picking_form`'s Operations tab; all `picking_type_code`-gated. |
| `static/src/js/stock_move_line_rows.js` | Shared row list (`getMoveLineRows`) all three columns read from. |
| `static/src/js/stock_move_line_batch_field.js` / `.xml` | Batch column: tags, delete, dispatches to inline dropdown (internal/outgoing) or popover (incoming). |
| `static/src/js/stock_move_line_batch_quickcreate.js` / `.xml` | Receipt-only "Add Batch" popover: live search + new-batch + expiry entry. |
| `static/src/js/stock_move_line_row_field.js` / `.xml` | Shared, parametrized widget powering both Batch Qty and Expiry Date columns. |
| `static/src/js/searchable_x2many.js` | Operations grid list renderer; also where the new-line auto-save (§8) lives. |
| `static/src/scss/stock_picking_operations.scss` | All styling for the above. |

No new Python model was added for this feature — it's view + JS/SCSS only, layered on top of Odoo's core `stock.picking` / `stock.move` / `stock.lot`.

## 8. Known trade-offs / open items

- **Demand quantity (Initial Demand) goes read-only shortly after a new line is added.** Side effect of auto-saving the picking as soon as a product is selected (needed so Batch/Batch Qty/Expiry become usable on that row at all) combined with core Odoo's own `_autoconfirm_picking()` / `is_initial_demand_editable` logic, which locks Demand once a move leaves `draft` state on an `is_locked` picking. Root-caused but not

yet resolved; the options on the table are (a) scope an override to just-added (`additional = True`) lines specifically, (b) move the auto-save trigger back to "leaving the row" instead of "product selected," or (c) unlock the whole picking (not recommended — that also reopens Scheduled Date editing picking-wide, wider than intended).

- **No warehouse/location scoping on batches** (§4) — a batch created on one GRN is selectable everywhere, company-wide, today. Unbuilt if the business actually wants batches scoped per warehouse/store.
- **"Higher official" governance (§6) is unbuilt** — line-item add/select/delete works for anyone today; editing an existing batch's own details has no extra gate yet.

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Internal implementation reference — Harleys Odoo 19 Enterprise. Click any screenshot to view it enlarged and step through the rest. Looking for a non-technical walkthrough instead? See the User Guide.